

# PyTurbo\_SF: An Adaptive Bootstrap Framework for Efficient Structure Function Analysis in Turbulent Flows

Adam Ayouche<sup>1</sup>, Baylor Fox-Kemper<sup>1</sup>, Momme Hell<sup>2</sup>, Brodie Pearson<sup>3</sup>, and Cassidy Wagner<sup>3</sup>

<sup>1</sup> Department of Earth, Environmental, and Planetary Sciences (DEEPS), Brown University, Providence, RI, USA <sup>2</sup> Woods Hole Oceanographic Institution, Woods Hole, MA, USA <sup>3</sup> College of Earth, Ocean, and Atmospheric Sciences, Oregon State University, Corvallis, OR, USA ¶ Corresponding author

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## Summary

Structure functions (SFs) are fundamental diagnostic tools in turbulence research that quantify spatial correlations between field differences at varying separation distances, revealing energy cascade characteristics and scaling laws (Frisch, 1995; Pope, 2000). PyTurbo\_SF is a comprehensive Python package providing efficient, statistically rigorous structure function calculations for 1D, 2D, and 3D turbulent datasets through an innovative adaptive bootstrap framework.

The package addresses computational bottlenecks through power-of-2 spacing strategies, adaptive convergence monitoring, and memory-optimized algorithms. PyTurbo\_SF delivers robust uncertainty quantification through bootstrap resampling and supports diverse (conditional) structure function types including longitudinal, transverse, scalar, advective, and energy flux functions (Pearson et al., 2021, 2024).

Applications span oceanographic time series and satellite measurements to high-resolution simulations, enabling consistent methodology across scales from laboratory to planetary systems. Despite its name evoking the broader turbulence field, PyTurbo\_SF is specifically designed for structure function analysis — one of the most widely used diagnostic approaches in turbulence research.

## Statement of need

Contemporary turbulence research relies on massive datasets from satellite missions, autonomous platforms, and high-resolution simulations. Traditional structure function calculations face severe limitations: computational intractability for large datasets, absence of uncertainty quantification, manual parameter tuning, and limited function types.

Existing tools address only subsets of these challenges. fastSF provides parallelized implementations but lacks advanced function types and uncertainty quantification (Sadhukhan et al., 2021). MATLAB toolkits are environmentally limited and lack comprehensive statistical frameworks (Fuchs et al., 2022). FluidSF (Wagner et al., 2025) is a related Python package supporting core structure function types, which we compare in detail below. Alternative approaches like FlowSieve cannot provide the scale-by-scale information structure functions uniquely deliver (Storer & Aluie, 2023).

There is a growing need for tools to analyze emerging datasets from NASA's SWOT satellite (Morrow et al., 2019) and next-generation atmospheric simulations generating terabyte-scale

40 outputs. Similarly, flux tower networks (e.g., AmeriFlux, FLUXNET) generate extensive eddy-  
 41 covariance time series of velocity and scalar turbulence that would benefit from standardized  
 42 structure function diagnostics with uncertainty quantification. Recent advances in structure  
 43 function theory, particularly advective structure functions (Pearson et al., 2021) and spectral  
 44 flux estimation (Pearson et al., 2024), require frameworks handling both traditional and novel  
 45 function types with statistical rigor.

46 PyTurbo\_SF fills this gap by providing the first comprehensive, statistically robust framework  
 47 that scales from small observational datasets to massive simulation outputs while delivering  
 48 quantified uncertainties essential for scientific interpretation.

## 49 Software functionality

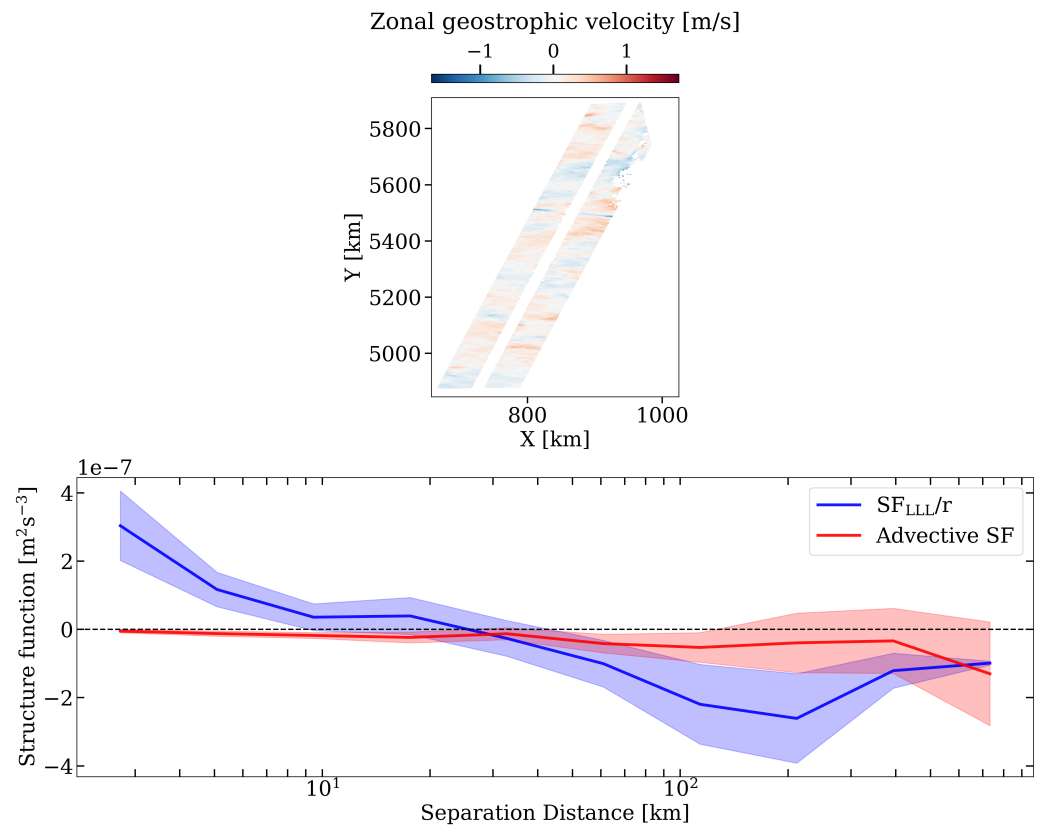
50 PyTurbo\_SF implements the complete mathematical framework for structure function analysis,  
 51 supporting functions of the form  $S_n(r) = \langle |\phi(\mathbf{x} + \mathbf{r}) - \phi(\mathbf{x})|^n \rangle_{\mathbf{x}}$  where  $\phi$  represents arbitrary  
 52 field variables (velocity, scalars, derived quantities),  $\mathbf{r}$  is the separation vector,  $n$  is the order,  
 53 and  $\langle \cdot \rangle_{\mathbf{x}}$  denotes spatial averaging. For velocity fields, the package computes **longitudinal** SFs  
 54 using the velocity component parallel to the separation vector ( $\delta u_L = \delta \mathbf{u} \cdot \hat{\mathbf{r}}$ ), **transverse** SFs  
 55 using the perpendicular component ( $\delta u_T = |\delta \mathbf{u} \times \hat{\mathbf{r}}|$ ), and **scalar** SFs for temperature, salinity,  
 56 or other tracers. Cross-structure functions correlate different components (e.g., longitudinal-  
 57 scalar), while **advective** SFs correlate velocity increments with advection increments to directly  
 58 quantify energy flux (Pearson et al., 2021).

59 The core algorithmic breakthrough is adaptive bootstrap methodology increasing computational  
 60 efficiency and statistical reliability. The algorithm employs power-of-2 spacings optimizing  
 61 memory access patterns while providing optimal scale separation. Adaptive convergence  
 62 monitoring dynamically allocates computational resources, eliminating manual parameter  
 63 tuning while guaranteeing robust uncertainty estimates.

64 Performance optimization enables analysis of previously intractable datasets. Memory-efficient  
 65 structures maintain peak usage at 2-5× base dataset size. Parallel processing provides near-  
 66 linear scaling with available cores. Benchmark testing demonstrates  $O(NM \log N \log M)$   
 67 complexity for 2D data, enabling analysis of datasets with millions of grid points.

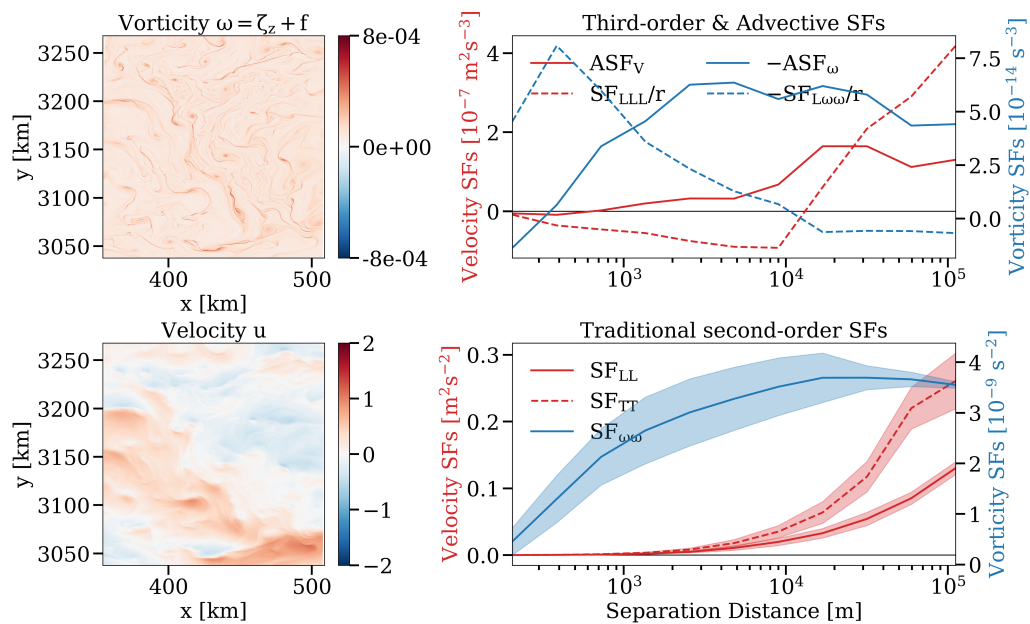
68 The package provides three main interfaces: `bin_sf_1d()` for time series, `bin_sf_2d()` for  
 69 surface fields, and `bin_sf_3d()` for volumetric data. All functions automatically optimize  
 70 computational strategies while supporting both isotropic and directional analysis. Applications  
 71 extend beyond traditional oceanographic and atmospheric modeling contexts to include eddy-  
 72 covariance field measurements, where high-frequency sonic anemometer time series of wind  
 73 components and scalars can be analyzed using `bin_sf_1d()` for temporal structure functions.

74 Figure 1 demonstrates PyTurbo\_SF applied to NASA SWOT satellite altimetry in the Gulf  
 75 Stream region. The third-order longitudinal and advective SFs reveal a sign change near 100  
 76 km separation, marking the transition from inverse to forward energy cascade characteristic of  
 77 quasi-geostrophic ocean turbulence.



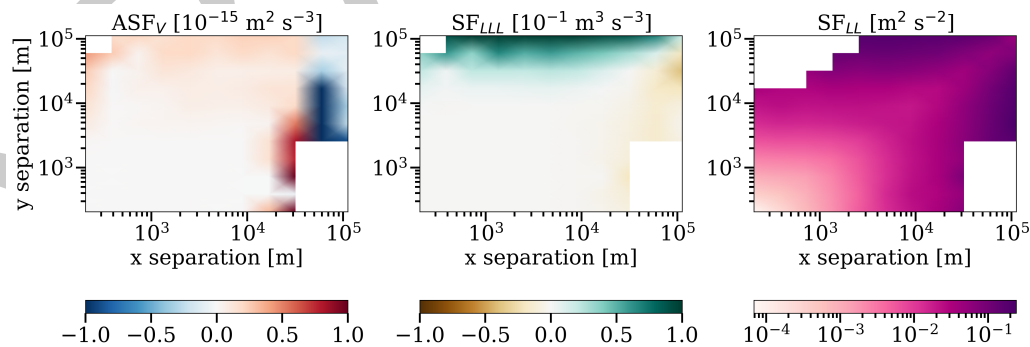
**Figure 1:** SWOT altimetry analysis in the Gulf Stream. **Upper:** zonal geostrophic velocity field ( $u_g$ , m/s). **Lower:** isotropic  $SF_{LLL}/r$  (blue) and advective SF  $ASF_V$  (red) in  $m^2 s^{-3}$ , with 95% bootstrap confidence intervals (shading). Positive values indicate forward cascade; negative values indicate inverse cascade.

78 Figure 2 applies the same framework to CROCO regional ocean model output (Shchepetkin  
79 & McWilliams, 2005). Both third-order energy-flux SFs and traditional second-order SFs are  
80 computed isotropically using `get_isotropic_sf_2d()`, with shaded 95% bootstrap confidence  
81 intervals on all quantities.



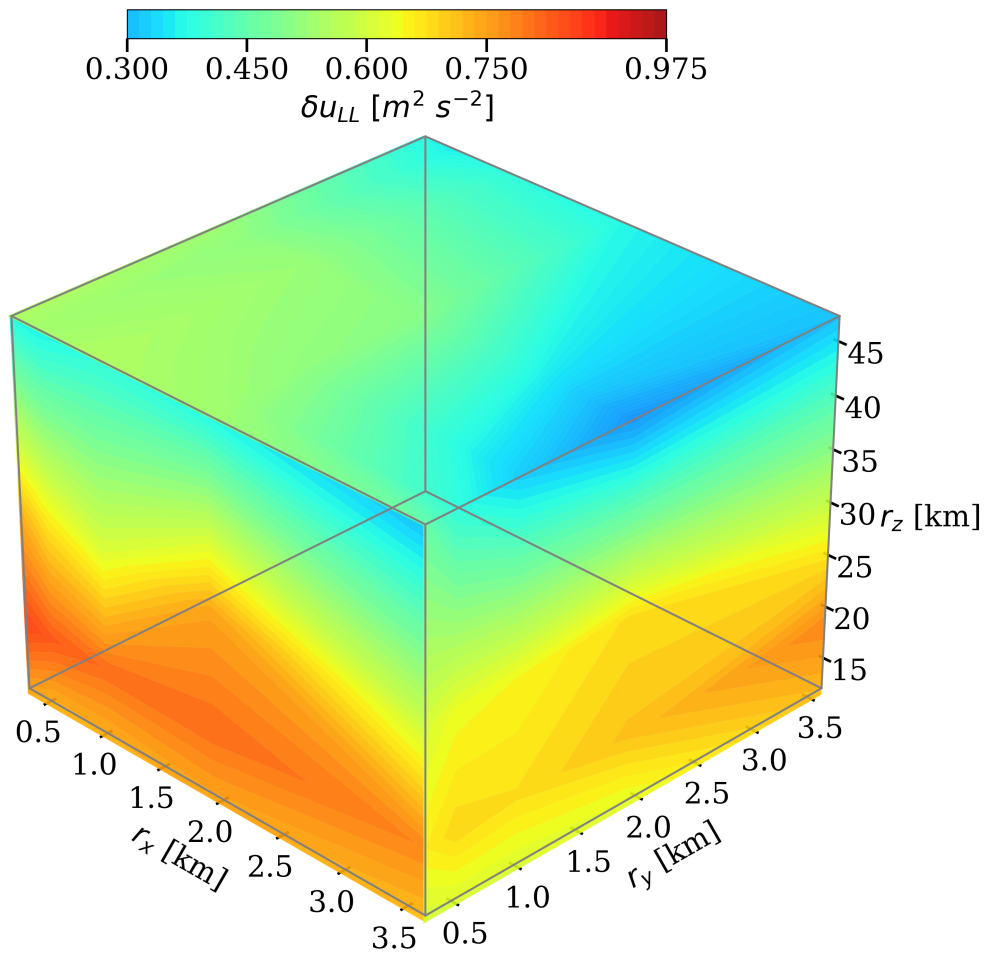
**Figure 2:** Isotropic SFs from CROCO simulation. **Left:** absolute vorticity  $\omega$  (upper) and zonal velocity  $u$  (lower) over a  $150 \text{ km} \times 250 \text{ km}$  domain. **Upper right:** third-order SFs ( $\text{ASF}_V$ ,  $\text{SF}_{LLL}/r$ ,  $-\text{ASF}_\omega$ ,  $-\text{SF}_{L\omega\omega}/r$ ). **Lower right:** second-order SFs ( $\text{SF}_{LL}$ ,  $\text{SF}_{TT}$ ,  $\text{SF}_{\omega\omega}$ ). Shaded regions: 95% bootstrap confidence intervals.

Figure 3 shows the full 2D (non-isotropic) SF output from the same CROCO snapshot, computed using `bin_sf_2d()`. The asymmetry about the diagonal confirms anisotropic energy transfer rates in zonal versus meridional directions.



**Figure 3:** Non-isotropic SFs from CROCO computed with `bin_sf_2d()`, showing SF amplitude as a function of zonal ( $x$ ) and meridional ( $y$ ) separation distance (log axes,  $10^3$ – $10^5$  m). **Left:**  $\text{ASF}_V$ . **Center:**  $\text{SF}_{LLL}$ . **Right:**  $\text{SF}_{LL}$ . White regions indicate insufficient data pairs.

Figure 4 illustrates the 3D capability using DYCOMS-II large eddy simulation output (Stevens et al., 2005). The second-order longitudinal SF computed with `bin_sf3d()` exhibits horizontal isotropy in the  $r_x$ - $r_y$  plane and distinct vertical anisotropy imposed by stable stratification at the boundary layer top.



**Figure 4:** 3D second-order longitudinal SF ( $SF_{LL}$ ,  $m^2 s^{-2}$ ) from DYCOMS-II LES, computed with `bin_sf_3d()`. Axes:  $r_x, r_y$  (horizontal, 0–3500 m) and  $r_z$  (vertical, 0–45000 m). Horizontal symmetry indicates isotropic turbulence; vertical anisotropy reflects stable stratification.

## Related work and scientific impact

PyTurbo\_SF represents a significant advancement by uniquely combining comprehensive function types, adaptive bootstrap methodology, and optimized algorithms. While fastSF provides basic parallelized calculations (Sadhukhan et al., 2021) and MATLAB toolkits offer specific analyses (Fuchs et al., 2022), no existing software delivers the combination of statistical rigor, efficiency, and breadth required for contemporary turbulence research.

FluidSF (Wagner et al., 2025) is a related Python package for structure function calculations, supporting 1D/2D/3D data and core structure function types including longitudinal, transverse, scalar, and advective functions. At the time of this writing, FluidSF provides second- and third-order structure functions with a simplified isotropic averaging approach and outputs results as NumPy arrays. It offers a velocity-scalar blended cross-term type and a single transverse decomposition in 3D.

PyTurbo\_SF is, to our knowledge, the first open-source package to combine arbitrary-order structure function computation, adaptive bootstrap uncertainty quantification with automatic convergence monitoring, and parallel processing in a single framework. It provides extensive cross-term structure function types (longitudinal-transverse, longitudinal-scalar, transverse-scalar, and scalar-scalar), full 3D transverse decomposition across all three coordinate planes (ij,

ik, jk), conditional structure functions, exact spherical and polar binning for isotropic averaging, and spectral energy flux estimation via Bessel transform — all accelerated by power-of-2 spacing strategies and joblib-based parallelization. Results are returned as richly annotated xarray Datasets with embedded metadata, convergence diagnostics, and quality masks.

PyTurbo\_SF's primary contributions are: (1) rigorous uncertainty quantification through adaptive bootstrap resampling with automatic convergence monitoring, (2) support for arbitrary-order structure functions essential for intermittency analysis, and (3) computational efficiency through parallelization and power-of-2 spacing strategies enabling analysis of large datasets.

The package enables application of recent theoretical developments, particularly advective structure functions providing direct energy flux measurements (Pearson et al., 2021) and spectral flux estimation methodologies (Pearson et al., 2024). These reveal energy pathways traditional approaches cannot capture, offering insights into cascade mechanisms in ocean and atmospheric turbulence.

Scientific applications demonstrate transformative impact across domains. PyTurbo\_SF enables analysis of satellite altimetry data for characterizing surface turbulence and large eddy simulation data for understanding boundary layer dynamics. The consistent methodology enables comparative studies previously impossible due to software limitations.

The adaptive bootstrap framework addresses a fundamental challenge: quantifying uncertainties in structure function estimates. PyTurbo\_SF's principled uncertainty quantification enables robust statistical comparisons and hypothesis testing, elevating scientific standards.

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